Redefining pancreatic cancer management with tumor-agnostic precision medicine.

PMID: 39514550 | Indexed in PubMed

Precision oncology and tumor-agnostic drug development provide hope for enhancing outcomes among patients with pancreatic cancer. Tumor-agnostic therapies have emerged across various tumor types, driven by insights into shared biomarkers. In the case of pancreatic cancer, the prevalence of the KRAS gene mutation is noteworthy. However, there exist other actionable alterations, such as BRCA1/2 mutations and fusion genes (BRAF, FGFR2, RET, NTRK, NRG1, and ALK), which present potential targets for therapy. Notably, tumor-agnostic drugs have demonstrated efficacy in specific subsets of pancreatic cancer patients who harbor these genetic alterations. Despite the rarity of NTRK fusions in pancreatic cancer, larotrectinib and entrectinib have exhibited effectiveness in NTRK fusion-positive pancreatic cancers. Additionally, repotrectinib, a next-generation NTRK inhibitor, has shown promising activity in NTRK positive pancreatic cancer patients who have developed acquired resistance to previous NTRK inhibitors. Immune checkpoint inhibitors, such as pembrolizumab and dostarlimab, have proven to be effective in dMMR/MSI-H pancreatic cancers. Moreover, targeted therapies for BRAF V600, RET fusions, and HER2/neu overexpression have displayed promising results in specific subsets of pancreatic cancer patients. Emerging targets like NRG fusions, FGFR2 fusions, TP53 mutations, and KRAS G12C mutations present potential avenues for targeted therapy. Tumor-agnostic therapies have the potential to revolutionize pancreatic cancer treatment by focusing on specific genetic alterations. It is crucial to continue implementing comprehensive screening strategies that encompass the ability to detect all these tumor-agnostic biomarkers. This will be essential in identifying pancreatic cancer patients who may benefit from these therapies.

Co-clinical Trial of Novel Bispecific Anti-HER2 Antibody Zanidatamab in Patient-Derived Xenografts.

PMID: 38358339 | Indexed in PubMed

Zanidatamab is a bispecific human epidermal growth factor receptor 2 (HER2)-targeted antibody that has demonstrated antitumor activity in a broad range of HER2-amplified/expressing solid tumors. We determined the antitumor activity of zanidatamab in patient-derived xenograft (PDX) models developed from pretreatment or postprogression biopsies on the first-in-human zanidatamab phase I study (NCT02892123). Of 36 tumors implanted, 19 PDX models were established (52.7% take rate) from 17 patients. Established PDXs represented a broad range of HER2-expressing cancers, and in vivo testing demonstrated an association between antitumor activity in PDXs and matched patients in 7 of 8 co-clinical models tested. We also identified amplification of MET as a potential mechanism of acquired resistance to zanidatamab and demonstrated that MET inhibitors have single-agent activity and can enhance zanidatamab activity in vitro and in vivo. These findings provide evidence that PDXs can be developed from pretreatment biopsies in clinical trials and may provide insight into mechanisms of resistance. We demonstrate that PDXs can be developed from pretreatment and postprogression biopsies in clinical trials and may represent a powerful preclinical tool. We identified amplification of MET as a potential mechanism of acquired resistance to the HER2 inhibitor zanidatamab and MET inhibitors alone and in combination as a therapeutic strategy. This article is featured in Selected Articles from This Issue, p. 695.

An anti-HER2 biparatopic antibody that induces unique HER2 clustering and complement-dependent cytotoxicity.

PMID: 36914633 | Indexed in PubMed

Human epidermal growth factor receptor 2 (HER2) is a receptor tyrosine kinase that plays an oncogenic role in breast, gastric and other solid tumors. However, anti-HER2 therapies are only currently approved for the treatment of breast and gastric/gastric esophageal junction cancers and treatment resistance remains a problem. Here, we engineer an anti-HER2 IgG1 bispecific, biparatopic antibody (Ab), zanidatamab, with unique and enhanced functionalities compared to both trastuzumab and the combination of trastuzumab plus pertuzumab (tras + pert). Zanidatamab binds adjacent HER2 molecules in trans and initiates distinct HER2 reorganization, as shown by polarized cell surface HER2 caps and large HER2 clusters, not observed with trastuzumab or tras + pert. Moreover, zanidatamab, but not trastuzumab nor tras + pert, elicit potent complement-dependent cytotoxicity (CDC) against high HER2-expressing tumor cells in vitro. Zanidatamab also mediates HER2 internalization and downregulation, inhibition of both cell signaling and tumor growth, antibody-dependent cellular cytotoxicity (ADCC) and phagocytosis (ADCP), and also shows superior in vivo antitumor activity compared to tras + pert in a HER2-expressing xenograft model. Collectively, we show that zanidatamab has multiple and distinct mechanisms of action derived from the structural effects of biparatopic HER2 engagement.

Zanidatamab, a novel bispecific antibody, for the treatment of locally advanced or metastatic HER2-expressing or HER2-amplified cancers: a phase 1, dose-escalation and expansion study.

PMID: 36400106 | Indexed in PubMed

HER2-targeted therapies have substantially improved outcomes for patients with HER2-positive breast and gastric or gastro-oesophageal junction cancers. Several other cancers exhibit HER2 expression or amplification, suggesting that HER2-targeted agents can have broader therapeutic impact. Zanidatamab is a humanised, bispecific monoclonal antibody directed against two non-overlapping domains of HER2. The aim of this study was to evaluate the safety and anti-tumour activity of zanidatamab across a range of solid tumours with HER2 expression or amplification. This first-in-human, multicentre, phase 1, dose-escalation and expansion trial included patients aged 18 years and older, with a life expectancy of at least 3 months, with an Eastern Cooperative Oncology Group performance status of 0 or 1, and locally advanced or metastatic, HER2-expressing or HER2-amplified solid tumours of any kind who had received all available approved therapies. The primary objectives of part 1 were to identify the maximum tolerated dose, optimal biological dose, or recommended dose of zanidatamab; all patients were included in the primary analyses. Part 1 followed a 3 + 3 dose-escalation design, including different intravenous doses (from 5 mg/kg to 30 mg/kg) and intervals (every 1, 2, or 3 weeks). The primary objective of part 2 was to evaluate the safety and tolerability of zanidatamab monotherapy in solid tumours. This trial is registered with ClinicalTrials.gov (NCT02892123), and parts 1 and 2 of the trial are complete. Part 3 of the study evaluates the use of zanidatamab in combination with chemotherapy and is ongoing. Recruitment took place between Sept 1, 2016, and March 13, 2021. In Part 1 (n=46), no dose-limiting toxicities were detected and the maximum tolerated dose was not reached. The recommended dose for part 2 (n=22 for biliary tract cancer; n=28 for colorectal cancer; and n=36 for other HER2-expressing or HER2-amplified cancers excluding breast or gastro-oesophageal cancers; total n=86) was 20 mg/kg every 2 weeks. The

most frequent treatment-related adverse events in part 1 of the study were diarrhoea (24 [52%] of 46 patients; all grade 1-2) and infusion reactions (20 [43%] of 46 patients; all grade 1-2). The most frequent treatment-related adverse events in part 2 of the study were diarrhoea (37 [43%] of 86 patients; all grade 1-2 except for one patient) and infusion reactions (29 [34%] of 86 patients; all grade 1-2). A total of six grade 3 treatment-related adverse events were reported in four (3%) of 132 patients. In part 2, 31 (37%; 95% CI 27·0-48·7) of 83 evaluable patients had a confirmed objective response. There were no treatment-related deaths. These results support that HER2 is an actionable target in various cancer histologies, including biliary tract cancer and colorectal cancer. Evaluation of zanidatamab continues in ongoing studies. Zymeworks.

Zanidatamab for HER2-amplified, unresectable, locally advanced or metastatic biliary tract cancer (HERIZON-BTC-01): a multicentre, single-arm, phase 2b study.

PMID: 37276871 | Indexed in PubMed

HER2 is overexpressed or amplified in a subset of biliary tract cancer. Zanidatamab, a bispecific antibody targeting two distinct HER2 epitopes, exhibited tolerability and preliminary anti-tumour activity in HER2-expressing or HER2 (also known as ERBB2)-amplified treatment-refractory biliary tract cancer. HERIZON-BTC-01 is a global, multicentre, single-arm, phase 2b trial of zanidatamab in patients with HER2-amplified, unresectable, locally advanced, or metastatic biliary tract cancer with disease progression on previous gemcitabine-based therapy, recruited at 32 clinical trial sites in nine countries in North America, South America, Asia, and Europe. Eligible patients were aged 18 years or older with HER2-amplified biliary tract cancer confirmed by in-situ hybridisation per central testing, at least one measurable target lesion per Response Evaluation Criteria in Solid Tumours (version 1.1), and an Eastern Cooperative Oncology Group performance status of 0 or 1. Patients were assigned into cohorts based on HER2 immunohistochemistry (IHC) score: cohort 1 (IHC 2+ or 3+; HER2-positive) and cohort 2 (IHC 0 or 1+). Patients received zanidatamab 20 mg/kg intravenously every 2 weeks. The primary endpoint was confirmed objective response rate in cohort 1 as assessed by independent central review. Anti-tumour activity and safety were assessed in all participants who received any dose of zanidatamab. This trial is registered with ClinicalTrials.gov, NCT04466891, is ongoing, and is closed to recruitment. Between Sept 15, 2020, and March 16, 2022, 87 patients were enrolled in HERIZON-BTC-01: 80 in cohort 1 (45 [56%] were female and 35 [44%] were male; 52 [65%] were Asian; median age was 64 years [IQR 58-70]) and seven in cohort 2 (five [71%] were male and two [29%] were female; five [71%] were Asian; median age was 62 years [IQR 58-77]). At the time of the data cutoff (Oct 10, 2022), 18 (21%) patients (17 in cohort 1 and one in cohort 2) were continuing to receive zanidatamab; 69 (79%) discontinued treatment (radiographic progression in 64 [74%] patients). The median duration of follow-up was 12·4 months (IQR 9·4-17·2). Confirmed objective responses by independent central review were observed in 33 patients in cohort 1 (41·3% [95% CI 30·4-52·8]). 16 (18%) patients had grade 3 treatment-related adverse events; the most common were diarrhoea (four [5%] patients) and decreased ejection fraction (three [3%] patients). There were no grade 4 treatment-related adverse events and no treatment-related deaths. Zanidatamab demonstrated meaningful clinical benefit with a manageable safety profile in patients with treatment-refractory, HER2-positive biliary tract cancer. These results support the potential of zanidatamab as a future treatment option in HER2-positive biliary tract cancer. Zymeworks, Jazz, and BeiGene.

A plain language summary of the results from the phase 2b HERIZON-BTC-01 study of zanidatamab in participants with HER2-amplified biliary tract cancer.

PMID: 39114870 | Indexed in PubMed

Researchers wanted to study whether the research drug zanidatamab could help people with a type of cancer called biliary tract cancer. In some people, biliary tract cancer cells make extra copies of a gene called HER2 (also called ERBB2). This is known as being HER2-amplified. Zanidatamab is an antibody designed to destroy cancer cells that have higher-than-normal HER2 protein or gene levels. Zanidatamab is currently under research and is not yet approved for any diseases. Participants in this phase 2b clinical study had tumors that were HER2-amplified and at the advanced or metastatic stage. Participants also had cancer which had become worse after previous chemotherapy or had side effects that were too bad to continue chemotherapy. They also had to meet other requirements to be enrolled. Researchers measured the amount of HER2 protein in the tumor samples of the participants who were enrolled. There were 80 participants with tumors that were both HER2 amplified and had higher-than-normal HER2 protein amounts (considered to be 'HER2-positive'). There were 7 participants with tumors that were HER2-amplified, but had little-to-no levels of the HER2 protein (considered to be 'HER2-low'). All participants in the study were treated with zanidatamab and no other cancer treatments once every 2 weeks. In the HER2-positive group, 33 of 80 (41%) participants had their tumors shrink by 30% or more of their original size. In half of these participants, their tumors did not grow for 13 months or longer. No participant in the HER2-low group had their tumors shrink by 30% or more. In total, 63 of 87 participants (72%) had at least one side effect believed to be related to zanidatamab treatment. Most side effects were mild or moderate in severity. No participant died from complications related to zanidatamab. Diarrhea was one of the more common side effects and was experienced by 32 of 87 participants (37%). Side effects related to receiving zanidatamab through the vein, such as chills, fever, or high blood pressure, were experienced by 29 of 87 participants (33%). The results of this study support the potential for zanidatamab as a new therapy for people with HER2-positive biliary tract cancer after they had already received chemotherapy. More research is occurring to support these results.Clinical Trial Registration: NCT04466891 (HERIZON-BTC-01 study).

A phase 2 trial of zanidatamab in HER2-overexpressed advanced endometrial carcinoma and carcinosarcoma (ZW25-IST-2).

PMID: 38262242 | Indexed in PubMed

HER2 overexpression is associated with decreased overall survival in metastatic endometrial cancer. Trastuzumab with chemotherapy has demonstrated efficacy for first-line management of advanced HER2+ endometrial carcinoma, but HER2-directed therapy in the recurrent setting is limited. Zanidatamab (ZW25), a humanized, bispecific antibody that simultaneously binds the 2 distinct HER2 epitopes bound by trastuzumab and pertuzumab, has demonstrated safety and activity in HER2+ tumors. Here, we report the results of a phase 2, open-label study evaluating the efficacy and safety of zanidatamab in patients with HER2+ metastatic endometrial carcinoma/carcinosarcoma who received prior treatment. We enrolled 16 patients with HER2+ endometrial carcinoma/carcinosarcoma after progression on ≤2 lines of therapy on a single-arm phase 2 study of zanidatamab. The primary endpoint was overall response rate (ORR; complete or partial response) by Response Evaluation Criteria in Solid Tumors version 1.1. HER2 immunohistochemistry and fluorescence in situ hybridization (FISH) were performed on pretreatment samples. Intratumor HER2 genetic heterogeneity was assessed. This study did not meet its primary efficacy endpoint. Although a clinical benefit rate of 37.5% was observed by 24 weeks, only 1 patient achieved a partial response (ORR, 6.2%). Eight patients had HER2 intratumor heterogeneity or lacked HER2 amplification by FISH. Decreased HER2 expression on repeat pretreatment samples was observed in 3 (75%) of 4 patients evaluated. We observed a low response rate to zanidatamab in recurrent HER2+ endometrial carcinoma/carcinosarcoma, which may be driven by downregulation of HER2 expression. Repeat HER2 testing should be considered prior to second-line HER2-directed therapy. govidentifier: NCT04513665.

HERIZON-GEA-01: Zanidatamab + chemo ± tislelizumab for 1L treatment of HER2-positive gastroesophageal adenocarcinoma.

PMID: 36000541 | Indexed in PubMed

HER2-positive gastroesophageal adenocarcinomas (GEAs) are common cancers with high mortality and the treatment options for advanced/metastatic disease are limited. Zanidatamab and tislelizumab are novel monoclonal antibodies targeting HER2 and PD-1, respectively, and have shown encouraging antitumor activity in early phase studies in multiple cancers, including GEA. Preliminary data suggest that dual targeting of the HER2 and PD-1 pathways could further improve upon the results achieved with targeting either pathway alone. Here, we describe the design of HERIZON-GEA-01, a global, randomized, open-label, active-comparator, Phase III study to evaluate and compare the efficacy and safety of zanidatamab plus chemotherapy with or without tislelizumab to the standard of care (trastuzumab plus chemotherapy) as first-line treatment for patients with advanced/metastatic HER2-positive GEAs.

Current and emerging therapies for advanced biliary tract cancers.

PMID: 34626563 | Indexed in PubMed

Biliary tract cancers (cholangiocarcinomas and gallbladder cancers) are increasing in incidence and have a poor prognosis. Most patients present with advanced disease, for which the treatment is palliative chemotherapy. Over the past few years, the genomic landscape of biliary tract cancers has been examined and several targeted therapies have been developed. Molecular targets with clinically meaningful activity include fibroblast growth factor receptor (FGFR), isocitrate dehydrogenase (IDH), RAS-RAF-MEK (MAP2K1)-ERK (MAPK3), HER2 (also known as ERBB2), DNA mismatch repair, and NTRK. Pemigatinib, a FGFR1-3 inhibitor, showed encouraging response rates and survival data as second-line treatment and received US Food and Drug Administration (FDA) approval in April, 2020, for previously treated advanced or metastatic cholangiocarcinoma with FGFR2 gene fusion or rearrangements. Ivosidenib, an IDH1 inhibitor, showed improved progression-free survival versus placebo in second-line treatment in the phase 3 ClarIDHy trial. Early phase trials of dabrafenib plus trametinib (BRAF and MEK inhibition) and zanidatamab (a bispecific HER2-antibody) have yielded encouraging response rates. Immunotherapy has mainly produced responses in tumours with deficient mismatch repair or high microsatellite instability (also known as dMMR or MSI-H) or higher PD-L1 score, or both. However, early phase trials of immunotherapy plus chemotherapy in unselected patient populations appear promising. NTRK inhibitors have also shown promise in early phase trials of NTRK-fusion positive solid tumours, including cholangiocarcinoma. In this Review, we discuss current and emerging therapies for advanced biliary tract cancers, with a focus on molecularly targeted therapy.

Revolutionizing anti-HER2 therapies for extrahepatic cholangiocarcinoma and gallbladder cancer: Current advancements and future perspectives.

PMID: 38266541 | Indexed in PubMed

Biliary tract cancers (BTCs) encompass a heterogeneous group of rare tumors, including intrahepatic cholangiocarcinoma (iCCA), extrahepatic cholangiocarcinoma (eCCA), gallbladder cancer (GBC) and ampullary cancer (AC). The present first-line palliative treatment regimen comprises gemcitabine and cisplatin in combination with immunotherapy based on two randomized controlled studies. Despite the thorough investigation of these palliative treatments, long-term survival remains low. Moving beyond conventional chemotherapies and immunotherapies, the realm of precision medicine has demonstrated remarkable efficacy in malignancies such as breast and gastric cancers, characterized by notable HER2 overexpression rates. In the context of biliary tract cancer, significant HER2 alterations are observed, particularly within eCCA and GBC, heightening the interest in precision medicine. Various anti-HER2 therapies, including trastuzumab, pertuzumab, trastuzumab-deruxtecan, zanidatamab and neratinib, have undergone investigation. The objective of this review is to summarize the current evidence and outline future directions of targeted HER2 treatment therapy in patients with biliary tract tumors, specially extrahepatic cholangiocarcinoma and gallbladder cancer.